

RMSE-Anchored Contrastive Calibration for Reliable Long-Tail Regression on Urban Proxy Dynamics

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Abstract

Urban proxy dynamics exhibit heavy-tailed distributions: most grids remain stable, while a small fraction of shocks dominate risk. Using CivitasGrid-TLP (2015–2024), we benchmark tabular regressors for grid-level prediction of $\Delta\text{WorldPop}$ and ΔVIIRS under `GroupSplit(grid_id)`. A two-expert Mixture-of-Experts (MoE) proves ineffective in this setting: tail routing is weakly identifiable, boundary decisions are unstable, and misrouting combined with inverse signed-log transformations amplifies errors, degrading both overall and tail metrics compared to a single XGBoost baseline. In contrast, a model-agnostic post-hoc affine calibrator trained with an RMSE-anchored contrastive objective consistently improves tail reliability while preserving overall fit across diverse baselines. The key insight is that in long-tail urban regression, reliable tail gains require anchoring—otherwise routing and calibration collapse.

I. INTRODUCTION

A. Economic Dynamics Prediction: Why This Matters

Transportation planning has become increasingly important, as effective transportation systems can boost local economies and enhance urban sustainability. In this context, developing tools that enable city planners to evaluate the potential economic outcomes of transportation infrastructure investments is critical. However, forecasting these economic outcomes remains challenging in practice. Economic changes result from complex interactions among regional conditions, municipal finances, and mobility patterns, making them difficult to predict with traditional modeling approaches. Machine learning models, which can capture complex nonlinear relationships, offer a promising alternative for economic dynamics modeling, though significant challenges remain.

B. The Challenge of Long-tail Urban Socioeconomic Dynamics Prediction

In this study, we construct and benchmark machine learning models to predict year-to-year changes in proxy urban economic signals (WorldPop and VIIRS) at the grid level.

We observe that differenced proxy signals follow a long-tailed distribution: most values cluster near zero, while a small but substantial number of grid cells experience abrupt shocks, placing them far from the bulk of the distribution. These outliers are driven by mechanistic factors that frequently occur in practice, rather than simple measurement noise, making it essential for models to reliably predict tail-grid outcomes. However, because most conventional machine learning models optimize for average error, they typically underperform when predicting tail dynamics. Heavy-tailed distributions can also cause training instabilities, degrade robustness under extreme events, and lead to larger-than-expected worst-case errors.

Large datasets are required for all machine learning models. However, official economic statistics are typically insufficient at fine spatial resolution, expensive to collect, temporally delayed, and inconsistent across regions. This motivates the use of scalable proxy signals for urban economic dynamics modeling.

C. Research Questions and Overview

These challenges motivate a core research question: how can we build a scalable and reliable machine learning framework for forecasting proxy-based urban economic dynamics under long-tailed targets and diverse regional conditions? To address this, we construct a grid-level dataset that integrates transportation network features with proxy economic signals, and we benchmark several strong baseline models under appropriate evaluation settings. We further investigate modeling strategies that optimize tail and worst-case performance, aiming to provide practical and robust tools for forecasting transportation infrastructure-driven economic outcomes.

D. Our Contributions

- **Dataset:** We construct a grid-level panel dataset integrating transportation network features with proxy economic signals (WorldPop / VIIRS), enabling scalable urban economic dynamics prediction.
- **Benchmark:** We provide a benchmark evaluation protocol and compare multiple strong baseline models under appropriate generalization settings for long-tail dynamics prediction.
- **Long-tail modeling:** We investigate modeling strategies that improve tail and worst-case performance under heavy-tailed targets, addressing the limitations of average-error-optimized regression models.
- **Practical relevance:** We frame the task as an infrastructure-driven economic outcome evaluation problem, aiming to support robust and practical decision tools for transportation planning.

Next, we review prior work on long-tail regression, mixture-of-experts learning under heterogeneity, and proxy-based socioeconomic indicators, which together motivate our benchmark setting.

II. LITERATURE REVIEW

A. Long-tail Regression

Traditional MSE/MAE-based regression models typically emphasize head (majority) samples at the expense of tail samples, resulting in reduced predictive accuracy on tail samples, particularly for heavy-tailed data. To address this bias, Ren et al. proposed a *Contrastive Balanced MSE* approach that improves model performance on long-tailed and imbalanced samples [1]. This method not only optimizes the distance between $\hat{y}_i$ and y_i , but also maximizes the distance between $\hat{y}_i$ and other samples' y_j , effectively reducing the dominant influence of head samples during training. Similarly, Yang et al. proposed *Deep Imbalanced Regression* [2], which combines *Label Distribution Smoothing* (LDS) and *Feature Distribution Smoothing* (FDS) to improve generalization under heavy-tailed targets. These complementary approaches motivate our setting, where long-tailed proxy dynamics interact with heterogeneous regional mechanisms and routing-based models.

B. Mixture-of-Experts

Yuksel et al. reviewed two decades of Mixture-of-Experts (MoE) research and applications [3]. Their work suggests that when data involve multiple regimes with heterogeneous mechanisms, a single model may be insufficient, but MoE with a soft routing gating network can substantially improve model fit by assigning each input to one or multiple experts. MoE can significantly improve fitting ability under heterogeneity; however, residual spikes may emerge when gating decisions are unstable.

While Yuksel et al. provide an overarching view of MoE as a divide-and-conquer framework, subsequent work formalizes MoE as a conditional mixture model. In Gormley and Frühwirth-Schnatter's research, MoE is formulated as a conditional mixture model where both mixing

weights (π_k) and expert distribution parameters depend on covariates [4]. MoE can model heterogeneity and subgroup structure effectively; however, the stability of MoEs depends critically on the smoothness and identifiability of the gating model.

Building on this statistical formulation, recent surveys highlight how MoE has been adapted to diverse learning tasks including regression under heterogeneous conditional distributions. Nguyen and Chamroukhi provide a comprehensive overview of MoE variants across regression, classification, and clustering [5], systematically reviewing hierarchical MoE, mixture of linear experts, time-series mixtures, and mixture-of-regression models with gating networks. Through this review, they emphasize that gating routing (soft vs. hard) and gating-model complexity influence stability and interpretability. In our work, we implement a hierarchical MoE approach that divides head and tail into two separate regimes, each handled by a mixture of two experts.

Beyond point prediction, recent work explores distributional regression using MoE-based approaches, which is particularly relevant for heavy-tailed targets. Rügamer et al. propose an innovative approach for processing long-tailed and heavy-tailed data with outliers and shocks [6]. Rather than predicting $\mathbb{E}[Y|X = x]$, distributional regression models predict the full conditional distribution $Y|X = x$, which significantly enhances robustness and reliability. However, in practice, long-tail regression benchmarks often use target transformations and gating-based experts, which can suffer from boundary-induced residual spikes due to uncertain routing and transformation inversion.

Overall, existing MoE literature establishes the framework as a powerful strategy for learning heterogeneous regimes, and recent distributional extensions highlight its potential for robustness under heavy-tailed targets. However, most studies focus on modeling flexibility, inference, and application-level performance, leaving a gap in systematically characterizing and mitigating stability issues in long-tail regression. Specifically, when gating decisions are uncertain near boundaries and expert predictions are limited under different target transformations, small differences can be amplified, leading to boundary-induced residual spikes and unacceptable worst-case errors. Motivated by this gap, our work empirically investigates stable MoE strategies under heavy-tailed proxy prediction and proposes an evaluation benchmark.

C. Proxy-based Socioeconomic Indicators

Beyond modeling strategies, our task depends on scalable, high-resolution proxies of economic dynamics, since official statistics are often unavailable at the grid level.

For population as an economic indicator, Paganelli notes that Adam Smith recognized population growth as a proxy for economic growth [7]: if a region can support population growth, its productivity is likely also increasing. Population change reflects aggregate outcomes of resource availability, labor demand, and wage dynamics rather than noise, providing meaningful signals for interpreting economic booms and recessions. This historical perspective validates our use of population-density data (WorldPop) when high-resolution official GDP data are unavailable. Peterson further emphasizes that population growth impacts vary by development level: in low-income economies it may dilute capital, while in high-income economies it can enable specialization [8]. This heterogeneity motivates training specialized experts for different development contexts.

Regarding nighttime luminosity as an economic proxy, Chen and Nordhaus demonstrate that nighttime light intensity is an objective, globally-available, spatially high-resolution indicator that can be used to replace or supplement traditional economic statistics [9]. Their systematic analysis of GDP and luminosity across 12,393 globally distributed $1\text{r} \times 1\text{r}$ grid cells (1992–2008) found a notably strong correlation ($R^2 = 0.97$) between illuminated area and GDP, validating VIIRS Nighttime Light Intensity as a proxy for economic dynamics. However, at finer spatial scales, the relationship becomes noisier and less reliable, especially

in low-output regions. Measurement error in log-luminosity across satellites introduces substantial uncertainty (SE 0.28–0.59), corresponding to $\geq 25\%$ proxy error. Thus, a more robust long-tail prediction model is essential.

Prior work primarily uses population or luminosity to approximate static economic output levels, but urban planning often requires predicting changes rather than absolute levels. This motivates defining targets as year-to-year differences rather than absolute values.

D. Literature Review Summary

In summary, prior work validates the use of population and nighttime luminosity as scalable economic proxies and extensively explores MoE for learning under heterogeneity. Long-tail regression research shows that standard MSE/MAE training biases models toward head samples, and remedies such as Balanced MSE and distribution smoothing (LDS/FDS) can improve tail performance under heavy-tailed targets. However, existing studies rarely provide unified benchmarks for grid-level proxy dynamics prediction integrating transportation morphology features, long-tailed targets, and robust evaluation. To address this gap, we introduce CivitasGrid-TLP as a benchmark dataset and investigate robust long-tail modeling strategies with controlled MoE and contrastive learning frameworks.

III. DATASET: CIVITASGRID-TLP

CivitasGrid-TLP is a grid-level panel dataset designed for proxy-based urban economic dynamics prediction [10]. It integrates transportation network features and remotely sensed proxy economic signals (WorldPop population density and VIIRS nighttime light) across thousands of city grids (30=60resolution).

A. Data Sources and Grid Definition

We define the basic spatial unit as a grid cell (30=60), and constructed a panel dataset spanning 20152024. Each grid-year record links a spatial cell with a calendar year, enabling year-to-year dynamics prediction.

Source	Type	Resolution	License
OpenStreetMap	Transportation	N/A	ODbL
VIIRS V2	Nighttime Light	15"	CC0
WorldPop	Population Density	1 km	CC BY

TABLE I: Data sources and specifications

As shown in Table I, we extracted transportation features from OpenStreetMap and linked them with proxy economic signals including VIIRS nighttime light intensity and WorldPop population density. Unlike existing work, which typically focuses on coarse administrative regions, CivitasGrid-TLP provides a grid-based benchmark at kilometer-level resolution, supporting robust evaluation across regions and countries.

B. Targets and Feature Groups

Model inputs in CivitasGrid-TLP are organized into three groups: (i) transportation network features, (ii) lagged proxy levels, and (iii) categorical descriptors. Transportation features are derived from OpenStreetMap (OSM) and aggregated into seven morphology categories capturing road density, hierarchy, and intersection structure. To ensure comparability across grids of different spatial resolutions, all OSM-derived features are normalized by grid area (per km²). When OSM coverage is absent or insufficient, transportation features are set to zero, indicating unmapped road infrastructure.

For proxy signals, we include previous-year absolute levels of WorldPop population density and VIIRS nighttime light intensity as minimal temporal context. The prediction targets

are year-to-year changes: $\Delta\text{WorldPop}$ and ΔVIIRS , which emphasize local dynamics while reducing baseline-level heterogeneity across regions.

Finally, categorical descriptors including country code, city type, and functional zone characterize heterogeneous regional contexts and enable stratified evaluation across development regimes.

C. Data Definition and Calculation

We define the prediction task as grid-level, year-to-year dynamics forecasting of proxy economic signals. Given features and lagged proxy observations for a grid cell, the goal is to predict proxy signal changes from year $t - 1$ to year t .

Specifically, we forecast two targets: population density change ($\Delta\text{WorldPop}$) and night-time light change (ΔVIIRS), defined as:

$$\Delta\text{WorldPop} = \text{WorldPop}_t - \text{WorldPop}_{t-1} \quad (1)$$

$$\Delta\text{VIIRS} = \text{VIIRS}_t - \text{VIIRS}_{t-1} \quad (2)$$

Differencing captures local economic dynamics and reduces baseline differences across regions. Most grid cells experience minor year-to-year fluctuations, while a non-negligible subset undergoes abrupt shocks. This long-tailed characteristic motivates strategies prioritizing tail and worst-case performance.

D. Quality Control and Filtering Tiers

Given substantial variation in data quality and completeness across regions and time, we adopt a tiered filtering procedure to construct subsets with different quality–coverage trade-offs. Transportation features are grouped into seven feature categories. For each grid-year record, we count available (non-missing) OSM feature categories.

We define two filtering tiers based on transportation feature completeness. Records with fewer than three available OSM feature categories are excluded due to insufficient information. Remaining records form the Medium-Quality (MQ) dataset. A stricter High-Quality (HQ) dataset is further defined by retaining records with at least four available OSM feature categories. This tiered design enables systematic evaluation under different quality constraints while maintaining transparency and reproducibility. We begin with 7,500 candidate grids. To focus on economically meaningful urban dynamics and exclude extremely sparse areas with limited analytical value, we first remove grids with population density below 5 km^{-2} . This leaves 6,178 unique grids in CivitasGrid-TLP v1.1.0. We then apply the MQ/HQ filtering criteria based on OSM feature completeness. The table below reports the resulting grid counts after each stage.

Dataset Tier	Grid Count
Initial Candidates	7,500
Population Masked	6,178
Medium-Quality (MQ)	4,293
High-Quality (HQ)	2,833

TABLE II: Dataset filtering results (grid counts)

In the next subsection, we report the resulting sample sizes, geographic coverage, and long-tail diagnostics for the MQ and HQ subsets.

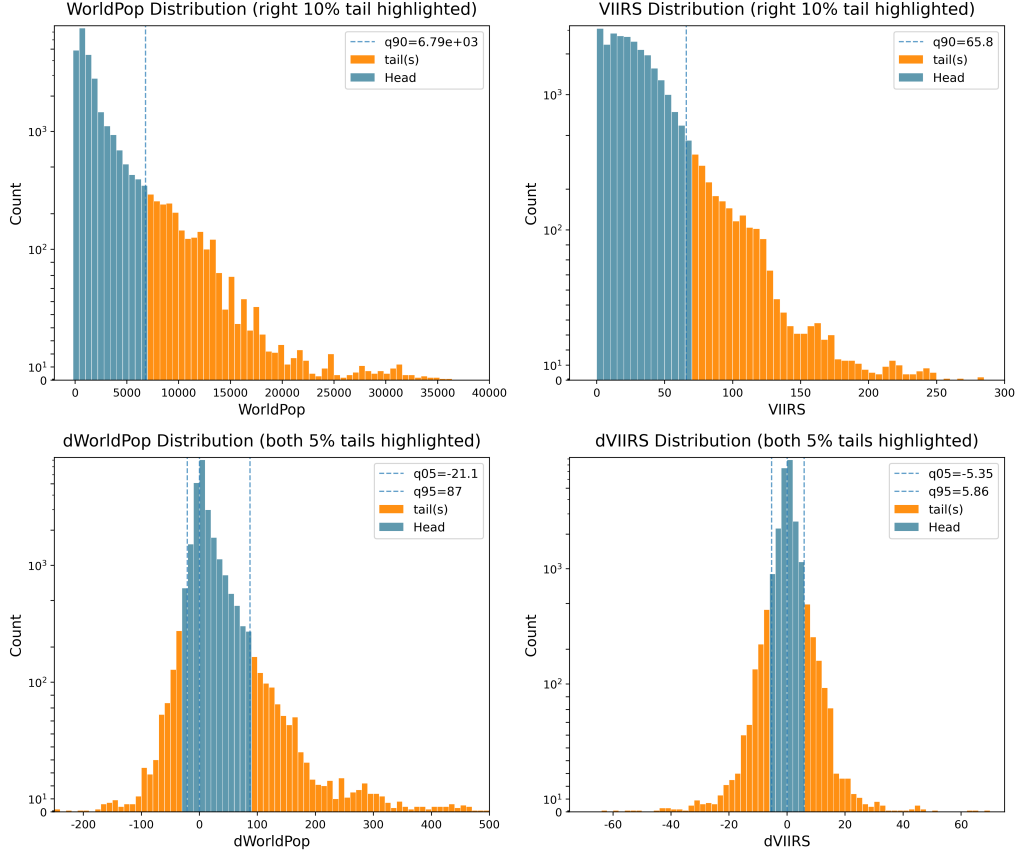


Fig. 1: Distribution of proxy signals and their year-to-year changes, illustrating the long-tailed nature of both absolute levels and changes. The y-axis uses a piecewise log scale: values below 100 are plotted linearly, while values above 100 use a logarithmic scale to reveal tail frequencies.

E. Dataset Statistics and Long-Tail Diagnostics

After removing candidate grids with population density below 5 km^{-2} , CivitasGrid-TLP v1.1.0 contains 6,178 unique grid cells observed over 10 years (2015–2024), yielding 61,780 grid-year records in total. Applying the OSM-completeness filters further retains 42,930 records in the Medium-Quality (MQ) subset and 28,330 records in the High-Quality (HQ) subset, corresponding to 4,293 and 2,833 unique grids, respectively.

The dataset covers 85 countries (HQ) or 100 countries (MQ) with diverse regional contexts, including nation code, region type, and city type as categorical descriptors. This diversity introduces heterogeneous feature distributions and motivates generalization evaluation across regions. Top 5 countries by grid count:

Rank	Nation Code	HQ Grid Count	MQ Grid Count
1	USA	1,715	2,523
2	CHN	401	581
3	IND	90	186
4	JPN	87	120
5	CAN	43	68
Total	N/A	2,833	4,293

TABLE III: Geographic distribution: top-5 countries by grid count

As shown in Table IV, both $\Delta\text{WorldPop}$ and ΔVIIRS exhibit high variance and strong

positive kurtosis, indicating heavy-tailed dynamics with frequent small fluctuations and occasional extreme shocks.

Target	Mean	Std. Dev.	Min	5%	25%	Median
$\Delta\text{WorldPop}$	16.37	52.54	-733.6	-21.10	-1.258	4.195
ΔVIIRS	0.190	4.333	-63.60	-5.354	-1.042	0.100

Target	75%	95%	Max	Skewness	Kurtosis
$\Delta\text{WorldPop}$	20.60	86.97	930.9	5.520	65.96
ΔVIIRS	1.402	5.864	68.86	0.174	29.41

TABLE IV: Target summary statistics

Figure 1 visualizes the distributions of proxy signals and their year-to-year changes for the HQ dataset. While absolute levels are right-skewed, differenced targets exhibit clear heavy-tailed patterns: most values concentrate near zero, while a substantial fraction lies far beyond. Such long-tailed targets pose challenges for average-error-optimized regression models.

Tail and Head Split. For each target y (ΔVIIRS and $\Delta\text{WorldPop}$), we define tail samples as grid-year records with absolute magnitude $|y|$ in the top 5% of the empirical distribution. This symmetric definition captures both positive and negative shocks, focusing on rare high-magnitude events. We use this tail subset throughout evaluations to characterize long-tail behavior and analyze model reliability under extreme events.

F. Spatial Coverage and Sampling Bias

Figure ?? shows the global spatial distribution of grids in CivitasGrid-TLP. Coverage is comprehensive but nonuniform: grids concentrate in regions with well-mapped road networks and stable proxy availability (North America, Europe, East and South Asia), while sparse in low-coverage areas. This imbalance is expected for datasets combining volunteered geographic information with remote sensing.

Non-uniform coverage introduces sampling through two channels. First, OSM completeness varies substantially across and within countries, affecting transportation feature availability. Second, quality-control filtering further excludes records with insufficient features, implicitly favoring well-mapped urbanized regions and disadvantaging rural areas. HQ/MQ subsets therefore over-represent documented regions while under-representing rural contexts.

These long-tail and coverage challenges motivate our modeling framework and evaluation protocol.

IV. METHODOLOGY

This section introduces the benchmark baselines and our controlled experimental variants, including an objective-level contrastive learning (CL) calibration layer and a two-expert MoE design for improving tail and worst-case performance.

A. Problem Setup and Tail Definition

We evaluate two strategies for improving tail performance—Mixture-of-Experts and contrastive learning calibration—on the CivitasGrid-TLP dataset. We consider a grid-level regression problem where each sample has features x_i representing year-to-year changes in proxy socioeconomic signals (WorldPop and VIIRS). The targets exhibit long-tailed distributions, with most samples near zero and a small but significant fraction at extreme

values. We define tail samples as those in the lower 5% or upper 5% of the empirical distribution:

$$\begin{aligned} i \in \mathcal{T}_{\text{VIIRS}} &\iff \begin{cases} \Delta\text{VIIRS}_i \leq -5.35, \\ \Delta\text{VIIRS}_i \geq 5.86, \end{cases} \\ i \in \mathcal{T}_{\text{WorldPop}} &\iff \begin{cases} \Delta\text{WorldPop}_i \leq -21.1, \\ \Delta\text{WorldPop}_i \geq 87. \end{cases} \end{aligned} \quad (3)$$

where the thresholds correspond to empirical percentiles. In CivitasGrid-TLP, each sample i is a *grid-year* record: a fixed spatial grid at year t , with features from transportation morphology, lagged proxy levels, and categorical descriptors, and targets as first differences $\Delta s_{i,t} = s_{i,t} - s_{i,t-1}$ for proxy signal s .

To ensure reproducibility and prevent leakage, all percentile thresholds are computed on the training split only and applied unchanged to validation and test splits.

We treat ΔVIIRS and $\Delta\text{WorldPop}$ as separate targets with independent tail sets. In evaluation, we report overall metrics on the full set and tail-specific metrics on the subset, since our goal is improving reliability under rare high-magnitude shocks.

B. Baseline Model

We choose XGBoost as our primary baseline because it performs well on tabular data through its ability to capture nonlinearities and feature interactions. The implementation is mature, efficient, and scalable, providing stability and interpretability.

XGBoost is a regularized gradient boosting framework that builds additive decision tree ensembles by iteratively fitting new trees to negative loss gradients using second-order Hessian information.

We also include Random Forest as a non-boosting classical baseline that provides complementary performance. Additionally, we include several standard baselines (Ridge Regression, Elastic Net, SVR, Extra Trees Regressor, and MLP) in controlled experiments to ensure our conclusions are not model-specific.

C. Contrastive Learning Evaluation

We evaluate *objective-level* contrastive learning (CL) as a generic tail-aware signal. Because many competitive tabular regressors are non-differentiable (e.g., random forests, SVR), we apply CL through a lightweight *post-hoc affine calibration* layer that can be trained on top of *any* baseline predictor.

Given a trained baseline $f(\cdot)$, we first obtain its predictions $\hat{y}^{(0)} = f(x)$. We then train an affine calibrator on the training split only:

$$\tilde{y} = a\hat{y}^{(0)} + b, \quad (4)$$

and evaluate $\tilde{y}$ on the held-out test split. We consider three configurations per baseline:

- **No-CL:** raw baseline prediction $\hat{y}^{(0)}$ (no calibrator);
- **CL-only:** calibrator trained using the contrastive objective $\mathcal{L}_{\text{con}}$, denoted as **(Baseline)_CL**;
- **RMSE+CL (mixed):** calibrator trained using a normalized mixture of RMSE and $\mathcal{L}_{\text{con}}$, denoted as **(Baseline)_RMSE_CL**.

This design enables a controlled comparison of objectives while keeping the underlying baseline model and data splits fixed.

1) *Contrastive objective:* Given a mini-batch $\mathcal{B}$, we treat the calibrated prediction $\tilde{y}_i$ as the anchor and its paired ground-truth y_i as the positive, while other y_j in the batch serve as in-batch negatives. Using an L1 distance $d(\tilde{y}, y) = |\tilde{y} - y|$ and temperature T , we define

$$p_i = \frac{\exp(-|\tilde{y}_i - y_i|/T)}{\sum_{j \in \mathcal{B}} \exp(-|\tilde{y}_i - y_j|/T)}, \quad (5)$$

and the contrastive loss

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} -\log p_i. \quad (6)$$

In our implementation, the temperature is set adaptively from the training target scale:

$$T = \max\left(\text{Std}(y_{\text{train}}) \cdot \kappa, 10^{-3}\right), \quad \kappa = 1.0. \quad (7)$$

2) *Normalized mixed objective (RMSE+CL)*: To combine RMSE with the contrastive loss in a numerically stable way, we optimize a normalized mixture on the calibrator parameters (a, b) :

$$\mathcal{L}_{\text{mix}} = w_{\text{rmse}} \cdot \frac{\text{RMSE}}{\text{RMSE}_0} + w_{\text{cl}} \cdot \frac{\mathcal{L}_{\text{con}}}{\mathcal{L}_{\text{con},0}}, \quad (8)$$

where $(w_{\text{rmse}}, w_{\text{cl}}) = (0.05, 0.95)$. The normalization constants are computed on the training split using the *initial* calibrator $(a, b) = (1, 0)$: RMSE_0 is computed exactly, while $\mathcal{L}_{\text{con},0}$ is estimated by subsampling at most 3000 training samples and averaging the batch-wise contrastive loss using a batch size of 512 to avoid an $O(N^2)$ full-dataset computation.

3) *Optimization details*: We optimize (a, b) with Adam (betas 0.9/0.999, $\epsilon = 10^{-8}$) for 40 epochs, batch size 128, and learning rate 0.05. This calibrator training is performed independently for each baseline and target using training-split predictions only, and then applied unchanged to validation/test predictions.

D. Two-expert MoE framework

As an architecture-level baseline to address heavy-tailed regression targets, we adopt a two-expert mixture-of-experts (MoE) design that blends a *body* regressor with a *tail* regressor through a learned soft gate. For a sample (x_i, y_i) , the MoE prediction is

$$\hat{y}_i = (1 - g(x_i)) \hat{y}^{(b)}(x_i) + g(x_i) \hat{y}^{(t)}(x_i), \quad (9)$$

where $\hat{y}^{(b)}(\cdot)$ and $\hat{y}^{(t)}(\cdot)$ denote the body and tail experts, and $g(x) \in (0, 1)$ is the gating probability. We implement all three components with XGBoost models, and use the same train/validation/test splits and preprocessing pipeline to ensure a fair comparison.

1) *Body Expert*: The body expert is a standard XGBoost regressor trained to predict the original target y :

$$\min_{\theta_b} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} \ell(y_i, \hat{y}^{(b)}(x_i; \theta_b)) + \Omega(\theta_b), \quad (10)$$

where $\Omega(\cdot)$ is the regularization term used by XGBoost. In our implementation, the body expert is trained on the full training split (rather than a filtered subset), with early stopping on the validation split when enabled.

2) *Tail Expert (signed-log target space)*: To stabilize extreme target magnitudes, the tail expert is trained in a signed-log space:

$$z = \text{signedlog1p}(y) = \text{sign}(y) \log(1 + |y|). \quad (11)$$

The tail expert is an XGBoost regressor trained to predict z on the full training split:

$$\min_{\theta_t} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} \ell(z_i, \hat{z}^{(t)}(x_i; \theta_t)) + \Omega(\theta_t), \quad (12)$$

where $z_i = \text{signedlog1p}(y_i)$. At inference time, we map predictions back to the original scale via the inverse transform

$$\hat{y}^{(t)} = \text{signedexpm1}(\hat{z}) = \text{sign}(\hat{z}) (\exp(|\hat{z}|) - 1). \quad (13)$$

3) *Gating model*: The gating model predicts whether a sample belongs to the tail region defined by training-split quantiles (Sec. IV-A). Let $(q_{0.05}, q_{0.95})$ be the 5th/95th percentiles computed on the training split. We define the tail indicator

$$r_i = \mathbb{I}(y_i \leq q_{0.05}) \vee \mathbb{I}(y_i \geq q_{0.95}), \quad (14)$$

and train an XGBoost binary classifier $g(x)$ by minimizing binary cross-entropy:

$$\min_{\phi} \sum_{i \in \mathcal{D}_{\text{train}}} \left[-r_i \log g(x_i; \phi) - (1 - r_i) \log(1 - g(x_i; \phi)) \right]. \quad (15)$$

At inference time, $g(x)$ provides a soft weight that smoothly blends body-like and tail-like behaviors without hard routing.

E. Evaluation protocol

This section presents technical details of our evaluation pipeline and metrics reported.

1) *Data Splitting*: We evaluate under two splitting strategies to assess both i.i.d. performance and spatial generalization robustness.

- 1) **Random Split (70/15/15)**: Randomly partition samples into 70% training, 15% validation, and 15% test.
- 2) **GroupSplit(grid_id)**: Split by grid_id to prevent spatial leakage: all samples from a grid belong to exactly one of train/val/test. Test grids are held out, and remaining grids are split into train and validation.

For each target, samples with missing labels are removed after splitting (filtering occurs within each split, per-target). All preprocessing (imputation, scaling, one-hot encoding) is fit on training data only and applied to validation/test.

2) *Metric Reporting*: We report standard regression metrics including MAE, RMSE, and R^2 . To summarize worst-case behavior, we additionally report high-quantile absolute error, namely the 95th and 99th percentiles of $|y - \hat{y}|$ on the evaluation set. For each method and each split, we compute metrics on the test set and report them with consistent prefixes for overall vs. tail evaluation (see below).

3) *Tail Evaluation*: Tail samples are defined as described in Sec. IV-A. In all experiments, the tail thresholds (e.g., quantiles) are computed using the training split only and then applied unchanged to validation/test to avoid leakage. We report the tail rate in the test split and compute all metrics on both: (i) the full test set (**All_**), and (ii) the tail subset (**Tail_**).

V. RESULTS

In this section we present and analyze experimental results across two targets and two splitting strategies. We find that contrastive learning (CL) calibration substantially improves tail performance, while the two-expert MoE approach underperforms the baseline.

A. Benchmark Performance (Baseline Models)

TABLE V: Benchmark performance on Δ VIIRS (dVIIRS)

Split	Model	R^2	RMSE	R^2_{tail}	RMSE _{tail}
RandomSplit	XGBoost	0.026	4.388	0.141	11.414
	Ridge	0.008	4.427	0.017	12.215
	ElasticNet	-0.000	4.446	-0.002	12.332
	SVR	-0.047	4.549	-0.014	12.403
	Random Forest	0.066	4.296	0.144	11.395
	ExtraTrees	0.014	4.414	0.131	11.485
	MLP	0.010	4.424	0.020	12.193
GroupSplit	XGBoost	0.058	4.675	0.137	12.869
	Ridge	0.025	4.757	0.030	13.648
	ElasticNet	-0.000	4.817	-0.000	13.856
	SVR	-0.009	4.838	0.022	13.706
	Random Forest	0.105	4.558	0.173	12.604
	ExtraTrees	0.093	4.587	0.168	12.639
	MLP	0.038	4.725	0.053	13.486

TABLE VI: Benchmark performance on Δ WorldPop (dWorldPop)

Split	Model	R^2	RMSE	R^2_{tail}	RMSE _{tail}
RandomSplit	XGBoost	0.836	21.976	0.832	61.048
	Ridge	0.354	43.593	0.366	118.483
	ElasticNet	0.274	46.197	0.215	131.789
	SVR	0.440	40.597	0.373	117.766
	Random Forest	0.784	25.209	0.770	71.417
	ExtraTrees	0.843	21.499	0.826	62.033
	MLP	0.766	26.239	0.791	68.011
GroupSplit	XGBoost	0.527	38.462	0.563	108.125
	Ridge	0.288	47.194	0.208	145.595
	ElasticNet	0.211	49.658	0.052	159.212
	SVR	0.374	44.234	0.283	138.528
	Random Forest	0.511	39.116	0.573	106.848
	ExtraTrees	0.466	40.875	0.513	114.195
	MLP	0.416	42.739	0.654	96.204

Benchmark results for both Δ VIIRS and Δ WorldPop are shown in Tables V and VI. Several patterns emerge:

First, predictions of Δ VIIRS (nighttime light change) are consistently weak across all baselines. Under random and group splits, R^2 values remain near or below zero, indicating minimal predictive signal beyond a mean baseline. This likely reflects noise amplification from the differencing procedure on an already noisy proxy. Thus, Δ WorldPop is treated as the primary target for evaluation.

Table VI shows Δ WorldPop benchmark performance under both Random Split (RS) and GroupSplit (GS) protocols, reporting overall (R^2 , RMSE) and tail-specific (R^2_{tail} , RMSE_{tail}) metrics. Under RS, tree-based models (ExtraTrees, XGBoost) achieve the strongest performance ($R^2 \approx 0.84$, RMSE ≈ 21 – 22), while MLP performs competitively and linear models lag. Under GS, all models degrade, reflecting increased difficulty in spatial generalization via grid_id splitting. Consistently, tail errors substantially exceed overall errors across all models.

Since RS may allow information leakage across years within the same grid_id, we treat **GS results as the primary reference** for subsequent comparisons. These benchmarks motivate focusing on tail robustness improvement under GS conditions.

B. Contrastive Learning Calibration: Main Results

We present and analyze Contrastive Learning-adjusted models compared to benchmarks. We compare Base (no calibration), Mix (RMSE+CL), and CL-only variants while keeping the

underlying predictor fixed. Both CL and RMSE+CL substantially improve tail performance over Base, but RMSE+CL maintains superior overall accuracy.

1) *How to Read Table VII*: The table is organized by target and split (rows) and by model family and calibration variant (columns). We report R^2_{all} , RMSE_{all} , R^2_{tail} , and $\text{RMSE}_{\text{tail}}$ for each configuration.

2) *Results and Interpretation*: Table VII reveals that CL calibration strongly depends on the objective. The mixed objective (RMSE+CL) provides consistent improvements, while CL-only is often unstable and substantially degrades overall fit.

Under GroupSplit on $\Delta\text{WorldPop}$ (our primary setting), RMSE+CL consistently improves tail robustness. For example, SVR shows the largest gains: R^2_{all} decreases slightly from 0.374 to 0.340, but R^2_{tail} rises significantly from 0.283 to 0.483, and tail RMSE improves from 138.528 to 117.573. Random Forest shows similar patterns: tail R^2 increases from 0.573 to 0.595 with $\text{RMSE}_{\text{tail}}$ improving from 106.848 to 104.055. XGBoost also benefits: R^2_{tail} improves from 0.563 to 0.577 and $\text{RMSE}_{\text{tail}}$ from 108.125 to 106.320. Tree ensembles like ETR, already strong baselines, show minimal response to calibration.

Strongest gains from CL-adjustment occur on tail metrics rather than overall metrics, confirming that the mixed objective acts as a tail-focused calibrator rather than global prediction rescaling—aligned with our objective of improving reliability on rare high-magnitude events.

In contrast, Table VIII shows CL-only frequently harms performance, suggesting contrastive objective alone is insufficient for preserving pointwise accuracy. Under GroupSplit, CL-only severely degrades Ridge and Elastic Net (both show negative overall R^2) and collapses SVR overall performance. This instability justifies using the RMSE-anchored mixed objective as default. Table VIII confirms this strategy remains stable and effective.

Overall, RS trends mirror GS patterns: RMSE+CL outperforms Base reliably, while CL-only is erratic. Since RS may allow temporal leakage within grid cells, we prioritize GS results as evidence of CL calibration effectiveness.

TABLE VIII: Average improvements of calibration variants over Base on **dWorldPop** under **GS**, aggregated across **seven models**. For R^2 , gain is $\text{Var} - \text{Base}$; for RMSE, gain is $\text{Base} - \text{Var}$ (so larger is better). Percentage gains are computed *per model* and then averaged.

Metric	Base (avg)	Gain (avg)	Gain (% avg)
Mix (RMSE+CL) vs. Base			
R^2_{all}	0.399	−0.0257	−8.11%
RMSE_{all}	43.183	−0.873	−1.94%
R^2_{tail}	0.407	+0.0763	+69.68%
$\text{RMSE}_{\text{tail}}$	124.101	+7.569	+5.37%
CL-only vs. Base			
R^2_{all}	0.399	−0.3176	−109.59%
RMSE_{all}	43.183	−9.334	−20.48%
R^2_{tail}	0.407	+0.0521	+40.17%
$\text{RMSE}_{\text{tail}}$	124.101	+5.208	+3.59%

C. Spike Diagnostics and Error Analysis

We investigate mechanisms underlying CL-adjustment improvements. Figure 2 visualizes CL calibration effects. After adjustment, tail samples (upper-right) move closer to the diagonal, indicating improved prediction accuracy. However, some body samples shift away, suggesting a tail–body trade-off. This visualization confirms CL-adjustment effectiveness for tail prediction while revealing refinement opportunities.

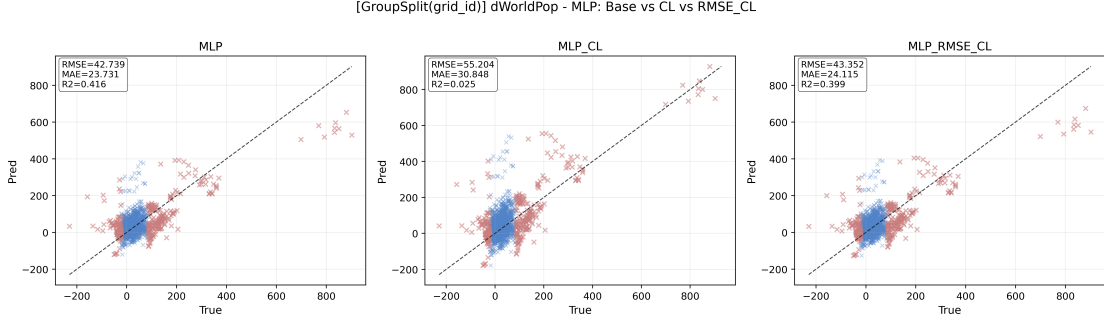


Fig. 2: Visualization of contrastive learning adjustment: a representative example showing how contrastive learning improves predictions for tail samples. In the middle graph (MLP_CL), an upper-right cluster of tail sample predictions improves substantially, while body predictions are stretched by the adjusting layer, leading to a decrease in overall performance.

With the mixed objective, overall performance is largely preserved while tail improves, showing that the RMSE component anchors predictions and prevents excessive body distortion.

Summary: CL calibration effectively improves tail predictions by reweighting model focus toward critical samples. The mixed objective maintains overall performance by balancing tail and body emphasis.

D. Ablations (CL Calibration)

We evaluate how mix-weight changes affect performance. Figure ?? shows the impact of varying mix weight on overall and tail metrics under GroupSplit on Δ WorldPop. Results indicate a balanced mix around 5% RMSE and 95

We also note that some CL-adjusted models (MLP, Random Forest, Extra Trees) show limited tail improvements. This likely reflects that these base models are already quite strong, leaving limited room for calibrator enhancement.

E. MoE: Negative Results

We attempted Mixture-of-Experts as an alternative architecture for long-tail regression. However, experiments demonstrate that the two-expert MoE underperforms the baseline.

Table IX shows that MoE (2-expert) under GroupSplit on Δ WorldPop degrades performance versus XGBoost across all metrics:

TABLE IX: MoE performance degradation on Δ WorldPop, GroupSplit(grid_id). For R^2 : gain = (MoE – Base)/Base $\times$ 100% (higher is better). For RMSE: gain = (Base – MoE)/Base $\times$ 100% (higher is better). Negative values indicate degradation.

Metric	Base	MoE	Gain (%)
Base (XGB) vs. MoE (2-expert)			
R^2_{all}	0.527	0.420	–20.30%
RMSE _{all}	38.462	46.474	–20.83%
R^2_{tail}	0.563	0.321	–42.98%
RMSE _{tail}	108.125	137.703	–27.35%

Thus, we investigate MoE’s underperformance through gating diagnostics in Figure ?. While the gating network differentiates many samples, it fails to achieve sufficient accuracy for expert specialization. This causes suboptimal or incorrect expert assignments, degrading overall performance.

VI. DISCUSSION

We evaluated machine learning models for predicting population and nighttime luminosity changes using CivitasGrid-TLP. The results demonstrate that RMSE-anchored contrastive calibration substantially improves tail performance without sacrificing overall accuracy. In contrast, the two-expert MoE unexpectedly underperformed. We also found that Δ VIIRS is substantially more challenging to predict than Δ WorldPop, likely reflecting increased noise from differencing. We discuss these findings in detail below.

A. Cross-Target Performance Differences

Δ WorldPop predictions consistently outperform Δ VIIRS across all models and metrics. Multiple factors explain this difference. First, nighttime light data generally contain more uncertainty and noise from atmospheric conditions, sensor degradation, and local lighting policies, whereas population data exhibit more stable patterns. Second, year-to-year differencing amplifies noise, which is particularly problematic for VIIRS because it is already noisy. Third, our transportation morphology features may be more informative for population changes than for light intensity changes, suggesting potential benefit from target-specific feature engineering.

B. RMSE+CL Calibration: Mechanisms and Evidence

The RMSE+CL mixed objective delivers consistent tail improvements. We attribute this to three mechanisms:

1) *Anchoring*: The RMSE component prevents the contrastive loss from over-emphasizing tail samples at the expense of body accuracy. By normalizing both terms and weighting them at 5% RMSE and 95% CL, the calibrator avoids the instability observed with CL-only (which frequently collapsed overall R^2 to negative values in Table VII).

2) *Dual Optimization*: RMSE minimization preserves pointwise accuracy while contrastive learning encourages separation between predictions and incorrect targets. This naturally pushes tail predictions (far from the mean) along the regression line, improving reliability without harming body predictions.

3) *Empirical Evidence*: Under GroupSplit on Δ WorldPop, RMSE+CL achieves a +69.68% average tail R^2 gain while changing overall RMSE by only -1.94% (Table VIII). This asymmetry shows effective rebalancing toward rare events without degradation. In contrast, CL-only sacrifices -109.59% overall R^2 for only $+40.17\%$ tail R^2 , indicating that pure contrastive learning is too aggressive.

C. MoE Failure Analysis

The two-expert MoE underperforms across both overall and tail metrics, highlighting challenges for routing-based approaches in long-tail settings.

First, predicting tail membership from available features is intrinsically difficult: the gating network struggles near quantile thresholds, where samples are inherently ambiguous. Small feature changes can flip expert assignments, amplifying variance and creating boundary artifacts. Second, misrouting is costly: routing a tail sample to the body expert (or vice versa) produces much larger errors than single-model baselines, which is especially problematic when rare extremes dominate tail metrics. Third, while the signed-log transformation stabilizes tail expert training, the inverse transformation magnifies small errors into large absolute values for extreme predictions.

These findings indicate that simple binary MoE with soft/hard gating is ill-suited for this task. More robust approaches—uncertainty-aware gating, continuous mixtures, or threshold-aware routing—may better handle the ambiguity and high stakes of long-tail routing.

D. Practical Recommendations

For long-tailed targets where tail reliability is critical, we recommend adding a post-hoc RMSE+CL calibration layer to strong baselines such as XGBoost or Random Forest. This lightweight, model-agnostic approach effectively enhances tail predictions while preserving overall performance, making it practical for practitioners.

E. Limitations and Future Work

We acknowledge several limitations and offer practical directions for future work:

- (1) *VIIRS* prediction remains unsatisfactory. Future work should explore smoothed targets (e.g., three-year trends) to reduce differencing noise and improve signal.
- (2) Our model predicts point estimates without uncertainty. Quantile regression or Bayesian approaches could enable uncertainty-aware prediction, improving decision support.
- (3) MoE’s failure may stem from weak gating networks. Future work should explore uncertainty-aware or threshold-informed routing mechanisms.
- (4) Spatial bias from OSM coverage differences could be mitigated through coverage-weighted training or post-hoc bias correction.

VII. CONCLUSION

Urban proxy dynamics are strongly heavy-tailed: most grid-year changes remain near zero, while rare shocks dominate error and risk. To study this regime reproducibly, we introduced CivitasGrid-TLP, a grid-level panel dataset (2015–2024) that integrates OSM transportation morphology with proxy socioeconomic signals (WorldPop and VIIRS). We evaluated models under both random splitting (potentially optimistic) and a leakage-resistant GroupSplit(grid_id) protocol.

Key findings across baselines: Δ WorldPop is substantially more predictable than Δ VIIRS, whose year-to-year differencing amplifies measurement noise, yielding near-zero R^2 for most methods. For long-tail robustness, our primary contribution is demonstrating that a lightweight, model-agnostic post-hoc affine calibrator trained with an *RMSE-anchored contrastive* objective consistently improves tail performance while preserving overall accuracy. Pure contrastive loss optimization is often unstable and severely degrades global fit. We also report a negative result: a two-expert MoE (body expert in original space, tail expert in signed-log space) underperforms a single XGBoost baseline on both overall and tail metrics, likely due to weak tail-routing identifiability and amplified transformation errors.

Takeaway: reliable tail gains require anchoring. For practitioners needing improved tail accuracy, RMSE-anchored contrastive calibration is a practical, lightweight add-on to strong tabular regressors. Naive tail/body MoE routing can collapse under feature–label ambiguity near quantile boundaries.

Future work should address VIIRS noise (via temporal smoothing or trend targets), incorporate uncertainty quantification (quantile or distributional regression), and explore uncertainty-aware or threshold-informed routing mechanisms to improve mixture-model stability in long-tail settings.

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CivitasGrid-TLP is publicly available at Zenodo (DOI 10.5281/zenodo.18144416) [10].

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REFERENCES

- [1] J. Ren, M. Zhang, C. Yu, and Z. Liu, “Balanced mse for imbalanced visual regression,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 7916–7925.
- [2] Y. Yang, K. Zha, Y. Chen, H. Wang, and D. Katabi, “Delving into deep imbalanced regression,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 11 842–11 851.
- [3] S. E. Yüksel, J. N. Wilson, and P. D. Gader, “Twenty years of mixture of experts,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 23, no. 8, pp. 1177–1193, 2012.
- [4] I. C. Gormley and S. Frühwirth-Schnatter, “Mixtures of experts models,” 2018, a chapter prepared for the forthcoming *Handbook of Mixture Analysis*.
- [5] H. D. Nguyen and F. Chamroukhi, “Practical and theoretical aspects of mixture-of-experts modeling: An overview,” *WIREs Data Mining and Knowledge Discovery*, vol. 8, p. e1246, 2018.
- [6] D. Rügamer, F. Pfisterer, B. Bischl, and B. Grün, “Mixture of experts distributional regression: implementation using robust estimation with adaptive first-order methods,” *AStA Advances in Statistical Analysis*, vol. 108, pp. 351–373, 2024.
- [7] M. P. Paganelli, “Population as a gdp proxy in adam smith,” SSRN working paper, 2021, forthcoming in *Journal of Scottish Philosophy*. [Online]. Available: <https://ssrn.com/abstract=3826675>
- [8] E. W. F. Peterson, “The role of population in economic growth,” *SAGE Open*, vol. 7, no. 4, pp. 1–15, 2017.
- [9] X. Chen and W. D. Nordhaus, “Using luminosity data as a proxy for economic statistics,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 108, no. 21, pp. 8589–8594, 2011.
- [10] W. Huang, “Civitasgrid-tlp (transluxpop): Grid-level transportation, nighttime lights, and population dynamics dataset,” Jan. 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.18144416>

TABLE VII: CL calibration results (Base vs. RMSE_CL (Mix) vs. CL) reported from `results_tail_tables.xlsx`. We report R^2_{all} /RMSE_{all} for overall (body) performance and R^2_{tail} /RMSE_{tail} for tail performance. Rows are grouped by target (dWorldPop, dVIIRS) and split (GS: `GroupSplit(grid_id)`, RS: `Random(70/15/15)`). Best value within each (Base, Mix, CL) triplet is marked with * at the displayed precision (ties starred).

Ridge Regression													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.288*	0.241	-0.336	47.194*	48.728	64.630	0.208	0.312*	0.253	145.595	135.709*	141.349
	RS	0.354*	0.323	-0.348	43.593*	44.623	62.960	0.366	0.447*	0.302	118.483	110.640*	124.307
dVIIRS	GS	0.025*	-0.054	-0.547	4.757*	4.945	5.991	0.030	0.035*	-0.104	13.648	13.613*	14.558
	RS	0.008*	-0.133	-0.738	4.427*	4.731	5.861	0.017*	-0.001	-0.151	12.215*	12.322	13.214
Elastic Net Regression													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.211*	0.177	-0.484	49.658*	50.727	68.117	0.052	0.239*	0.148	159.212	142.664*	150.966
	RS	0.274*	0.266	-0.472	46.197*	46.458	65.804	0.215	0.412*	0.237	131.789	114.091*	129.991
dVIIRS	GS	-0.000*	-0.006	-0.059	4.817*	4.831	4.956	-0.000*	-0.001	-0.007	13.856*	13.861	13.905
	RS	-0.000*	-0.009	-0.090	4.446*	4.464	4.641	-0.002	-0.000*	-0.004	12.332	12.319*	12.344
Support Vector Regression (SVR)													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.374*	0.340	-0.007	44.234*	45.428	56.116	0.283	0.483*	0.473	138.528	117.573*	118.693
	RS	0.440	0.511*	0.260	40.597	37.910*	46.664	0.373	0.551*	0.485	117.766	99.641*	106.807
dVIIRS	GS	-0.009*	-0.326	-1.743	4.838*	5.547	7.977	0.022*	-0.090	-0.708	13.706*	14.469	18.108
	RS	-0.047*	-0.402	-1.755	4.549*	5.263	7.378	-0.014*	-0.164	-0.771	12.403*	13.287	16.392
Random Forest Regressor (RF)													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.511*	0.504	0.480	39.116*	39.373	40.327	0.573	0.595	0.622*	106.848	104.055	100.549*
	RS	0.784	0.786*	0.776	25.209	25.087*	25.697	0.770	0.776*	0.769	71.417	70.385*	71.537
dVIIRS	GS	0.105*	0.048	-0.041	4.558*	4.700	4.914	0.173	0.203	0.204*	12.604	12.372	12.365*
	RS	0.066*	-0.014	-0.090	4.296*	4.476	4.641	0.144	0.157*	0.149	11.395	11.313*	11.365
Extra Trees Regressor (ETR)													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.466*	0.465	0.466*	40.875	40.889	40.861*	0.513	0.514*	0.511	114.195	114.012*	114.409
	RS	0.843*	0.843*	0.843*	21.499*	21.502	21.510	0.826*	0.826*	0.826*	62.033*	62.045	62.066
dVIIRS	GS	0.093	0.094*	0.088	4.587	4.586*	4.601	0.168	0.168	0.171*	12.639	12.642	12.619*
	RS	0.014*	0.011	0.009	4.414*	4.420	4.425	0.131*	0.131*	0.131*	11.485	11.481	11.480*
Multi-layer Perceptron Regressor (MLP)													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.416*	0.360	0.025	42.739*	44.742	55.204	0.654	0.660*	0.576	96.204	95.393*	106.494
	RS	0.766*	0.756	0.435	26.239*	26.766	40.760	0.791*	0.790	0.507	68.011*	68.178	104.449
dVIIRS	GS	0.038*	-0.094	-1.301	4.725*	5.039	7.307	0.053*	0.037	-0.472	13.486*	13.596	16.813
	RS	0.010*	-0.168	-0.934	4.424*	4.805	6.182	0.020*	0.006	-0.181	12.193*	12.282	13.385
XGBoost													
Target	Split	R^2_{all}			RMSE $_{\text{all}}$			R^2_{tail}			RMSE $_{\text{tail}}$		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.527*	0.526	0.426	38.462*	38.499	42.359	0.563	0.577	0.628*	108.125	106.320	99.790*
	RS	0.820	0.823*	0.666	23.018	22.821*	31.368	0.822	0.827*	0.672	62.821	61.846*	85.215
dVIIRS	GS	0.058*	0.003	-1.018	4.675*	4.810	6.843	0.137	0.143*	-0.152	12.869	12.825*	14.872
	RS	0.050*	0.002	-1.136	4.333*	4.441	6.496	0.144*	0.140	-0.315	11.398*	11.421	14.125

VIII. APPENDIX

TABLE X: Hyperparameters used in Contrastive Learning experiments

Component	Hyperparameter	Value
Global	Random seed	SEED=42
	Splitting ratios	Train/Val/Test = 70% / 15% / 15%
	Group split	GroupShuffleSplit; two-stage: hold out test groups, then split remaining groups into train/val
Tail (for reporting)	Tail quantiles	q05=0.05, q95=0.95 (computed on train split only)
	Tail mask	$y \leq q_{0.05}$ or $y \geq q_{0.95}$
Baselines (sklearn)	Ridge	$\alpha = 1.0$
	ElasticNet	$\alpha = 1.0$, l1_ratio=0.5, max_iter=5000, random_state=42
	SVR (RBF)	$C = 10.0$, $\epsilon = 0.1$, kernel=rbf
	RandomForestRegressor	n_estimators=150, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1
	ExtraTreesRegressor	n_estimators=300, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1
	MLPRegressor	hidden_layer_sizes=(128,64), activation=relu, early_stopping=True, validation_fraction=0.1, n_iter_no_change=15
CL calibrator (affine)	Calibrator form	$\tilde{y} = a\hat{y} + b$ (trainable a,b)
	Mode	cl (contrastive-only) or rmse_cl (mixed)
	CL batch size	CL_BATCH_SIZE=128
	Epochs	CL_EPOCHS=40
	Learning rate	CL_LR=0.05
	Temperature	$T = \max(\text{Std}(y_{\text{train}}) \cdot \text{CL_TEMP_FACTOR}, 10^{-3})$, CL_TEMP_FACTOR=1.0
	Numerical eps (loss)	CL_EPS=1e-12
	Mixed-loss weights	rmse_weight=0.05, cl_weight=0.95
	Normalization (rmse0)	rmse0 = RMSE on train with initial (a,b)=(1,0)
	Normalization (cl0)	Subsample estimate: CLO_MAX_N=3000, CLO_BATCH=512
	Adam update eps	eps=1e-8